SUPPLIER ONBOARDING AGENT

From one email to a protected supplier, step by step

We follow one real email from the demo, Al Noor Trading's, through every step the system takes. Then we look at what changes when something is wrong, how the agent keeps guarding the supplier afterwards, and how we tested all of it.

Who's involved

Every step below is done by one of these. The coloured labels show up next to each step.

Supplier **Faisal at Al Noor**
The company that wants to start selling to us.

Agent **The coordinator**
An AI that decides which step to do next and writes the emails. It can only act through a fixed set of buttons ("tools").

Reader **The document reader**
An AI that looks at each PDF and writes down the facts on it. Nothing else.

Checker **Plain code rules**
Ordinary code, no AI. Does the checks that must be exactly right, every time.

Registries **Wathq and ZATCA**
Government records of companies and VAT. Simulated in our demo.

You **The manager**
Only called in when something looks like fraud.

Part 1 · Al Noor's journey, start to finish

This is what really happened in the demo, with the real data. In real time it takes about 15 seconds and costs about 7 cents of AI.

1

Supplier **Faisal sends an email with 4 PDFs**

No portal, no account, no form. Just a normal email to the onboarding address.

THE EMAIL

```
From: Faisal Al-Harbi <faisal@alnoor-trading.com.sa>
Subject: New supplier onboarding - Al Noor Trading Co.

Dear Procurement Team,
Please find attached our company documents for supplier registration.
Best regards, Faisal Al-Harbi · Al Noor Trading Co. · +966594821993
```

سجل_تجاري.pdf | tax_cert.pdf | bank.pdf | nitaqat_gosi.pdf

Notice the file names don't say which document is which. One is even named in Arabic. The system has to work that out itself.

2

Checker

The system opens a case file

The email is saved, the 4 PDFs are stored, and a new case is created with the status “received”. The first line goes into the log: *Email from faisal@alnoor-trading.com.sa (4 attachments)*. From here on, every single action is written to that log. That is what the dashboard's right-hand panel shows.

3

Agent

The agent wakes up and opens the case

It sees the sender, the subject, the email text and the list of 4 unread documents. The email text is wrapped in a label that says *untrusted*: anything written in it is information, never an order the agent should follow.

4

Agent Reader

The 4 PDFs are read, all at once

The agent asks the Reader to read the documents. The Reader looks at each PDF like a person would and fills in the same short form for every one. This is what came back:

WHAT THE READER WROTE DOWN

File	It's a...	Company	Key number	Expires
سجل_تجاري.pdf	Commercial registration, **in Arabic**	شركة النور للتجارة	CR 2050260181	2027-10-20
tax_cert.pdf	VAT certificate	Al Noor Trading Co.	VAT 359083016613183	2028-03-14
bank.pdf	Bank letter	Holder: Al Noor Trading Co.	IBAN SA40 8009 …4628	—
nitaqat_gosi.pdf	GOSI certificate, in Arabic	شركة النور للتجارة	CR 2050260181	2028-01-09

The Reader only reads. It does not decide whether anything is OK. It also says whether the document is legible, and whether it contains any text trying to give instructions to an AI.

5

Checker

Code tidies up the facts

Small things that must be exact are fixed by code, not left to the AI: Arabic-Indic digits like ٢٠٥٠ become 2050, Hijri dates (like 1449/03/15 AH) are converted to the normal calendar with a proper formula, and spaces are stripped from IBANs.

6

Agent Checker Registries

17 checks are run

The agent asks the Checker to check everything. These are plain code rules from the settings file, so the answer is the same every time. For Al Noor, all 17 passed:

THE 17 CHECKS

✓ All 4 documents present

- CR, VAT, bank letter, GOSI

✓ Nothing expired (3)

- CR valid to 2027-10-20
- VAT to 2028-03-14
- GOSI to 2028-01-09

✓ Registries (3)

- Wathq: CR is **active**
- Name on CR matches Wathq
- ZATCA: VAT is registered

✓ IBAN is valid (1)

- Check digits add up, Saudi bank code 80 (Al Rajhi)

✓ Everything matches (6)

- Bank account holder = the company on the CR
- VAT and GOSI names match the CR
- Same CR number on all documents

The Arabic name شركة النور للتجارة and the English name Al Noor Trading Co. are matched through the registry, which lists both.

7

Checker

The verdict: all clear

✓ Recommended action: onboard

There are only three possible verdicts: **all clear**, **supplier can fix it**, or **risk, a human must decide**. Part 2 shows the other two.

8

Agent Checker

The agent creates the vendor record

The agent presses the “create vendor” button. The record is saved as `V-2050260181` with the company name in both languages, CR and VAT numbers, the IBAN, the email domain, the phone number from the email, and every document's expiry date.

> **The lock:** that button re-runs all the checks itself before doing anything. If even one check fails, it refuses, whatever the agent was told and whatever the documents say. The AI cannot talk its way past this.

9

Agent The agent writes and sends a welcome email

THE EMAIL IT WROTE

```
To: faisal@alnoor-trading.com.sa
Subject: Welcome - Al Noor Trading Co. onboarded

Dear Faisal, thank you for your documents. All checks passed and Al Noor
Trading Co. is now registered as a vendor (ID V-2050260181). We look forward
to working with you. Best regards, Procurement Team
```

Emails can only ever go back to the address that opened the case.

✓

Agent Case closed: onboarded. No human needed.

The agent marks the case done. It used 6 actions out of the 12 it is allowed per turn. Al Noor is now a vendor, and from this moment the system starts **guarding** it (Part 3).

Part 2 · When something is wrong

Steps 1 to 6 are always the same. What changes is the verdict at step 7.

Supplier can fix it

Rawabi Logistics: VAT certificate expired

1. Check “nothing expired” fails for the VAT certificate.
2. The agent emails Rawabi asking for **exactly** that document, and says when it expired.
3. Case status: *waiting on supplier*. No human involved.
4. Rawabi replies with a renewed certificate.
5. Only the new PDF is read; the 17 checks run again; all clear.
6. Vendor created, welcome email sent.

If the same problem comes back after 2 emails, it goes to a human.

Risk: a human decides

Gulf Steel: bank account in another company's name

1. Check “bank account holder = company on the CR” fails. The letter says *Desert Bloom Trading Co.*
2. This is a fraud pattern, so the agent is **not allowed** to email the supplier about it. The email button refuses.
3. It raises a red decision card for you, with the evidence.
4. You choose: ask the supplier to explain, approve anyway, or reject.

The same happens for an inactive registration, mismatched numbers, or a document that tries to give the AI orders.

Double-check first

A number that doesn't add up

1. On a blurry scan, the Reader misreads one digit of a CR number.
2. The registry can't find it, and it disagrees with the other documents.
3. Before raising any alarm, the code sends that one document back for a slow, careful second read.
4. The corrected number matches, and the case continues normally.

A real mismatch survives the second read; a misread digit does not.

Part 3 · Guarding the supplier afterwards

Most tools stop at step 10. This one keeps watching.

A

Fraudster

A fake email asks to change Al Noor's bank

It comes from `faisal@alnoor-trading.co`. The real domain is `alnoor-trading.com.sa`. It asks to send all future payments to a new IBAN at Bank AlJazira, and says it's urgent.

B

Agent

The agent recognises a bank-change request for Al Noor

It reads the new IBAN and account name from the email and passes them to the bank guard.

C

Checker

The bank guard runs 4 exact checks

RESULT	
Is the IBAN valid?	Yes
Same bank as the letter on file?	No: AlJazira instead of Al Rajhi
Account holder is Al Noor?	Yes
Sent from the domain on file?	No: alnoor-trading.co

D

Agent

The change is frozen and payments are held

The agent has no button that changes bank details. The only one it has is **freeze**. Payments to Al Noor go on hold, and a red card appears for you: *"Call +966594821993, the number on file, before approving."* It does not reply to the sender, because that would be talking to the fraudster.

Even a genuine request with zero red flags gets frozen. Every bank change needs a phone call.

E

You

You call Al Noor. They never asked for this. You reject it.

Payments are released and the bank details stay as they were. Only a person can approve a bank change.

F

`Checker` **Every day, all document dates are checked**

In the demo the clock jumps 11 months to 30 Sep 2027. The daily check finds:

- **Al Noor's CR expires in 20 days.** A renewal email goes out automatically and Al Noor is marked *at risk*.
- **Rawabi's GOSI certificate has already expired.** Payments to Rawabi go on hold until they send a new one.

Part 4 · How we proved it works

The building and testing, start to finish.

1

Make realistic fake suppliers, with the right answers attached

A program generates companies, real-looking PDFs (some Arabic, some scanned and blurry, some with Hijri dates, some with hidden fraud) and the emails. For each case it also writes down the correct decision, so we can mark the agent like an exam.

2

Practice on 57 cases

First score: 96.5% correct, 100% of fraud stopped.

3

Fix mistakes one at a time, and write each one down

Four fixes (date formats, Hijri vs normal years, the careful second read, the IBAN check) took practice to 100%. Every case that ever failed was added to an automatic test, so it can't quietly break again.

4

Compare against simpler versions

No AI at all (rules only), AI only (no code checks), no registry, no cross-checking. This shows what each part adds.

5

Stress tests

A new rule (“Zakat certificate required”) added in the settings file with no code changes: still 100%. Three runs from scratch: 100%, 100%, 100%.

6

Freeze, then sit the exam once

The system was locked at a fixed version. Then 57 brand-new cases with document layouts it had never seen were generated and run once: **100% correct, 100% of fraud stopped**. AI only scored 98.2%, no AI scored 52.6%.

✓

Everything saved, so replaying costs nothing

Every AI answer is stored. The demo and every result can be rerun for free, without an API key, and give the same numbers.

The whole thing in seven lines

1. A supplier emails their documents.
2. An AI reads each PDF and writes down the facts.
3. Plain code checks those facts against the rules and the government registries.
4. All clear: the agent creates the vendor. Fixable: it emails the supplier. Risky: it asks you.
5. Locks in the code stop the AI from approving anything that failed a check, or from changing bank details.
6. Afterwards it freezes any bank-change request until you phone the supplier, and chases expiring documents.
7. We tested it on 57 unseen cases, once, after locking it: 100% correct, 100% of fraud stopped.